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ORIGINAL PAPER

Trends in transfusion-transmissible infections in potential donors in a specialist hospital in Rivers State

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Background and Objectives Transfusion-transmissible infections are still a major concern in developing countries because of the endemicity of these infections in the general population. Blood donors can be a source of continued transmission of these infections. This study determines the trends and prevalence of transfusion-transmissible infections amongst prospective donors.

Materials and Methods Screening documentation for prospective donors from 2011 to 2013 was reviewed; age, sex and type of donor were identified.

Results A total of 5600 prospective donors were screened over a 3-year period (2011–2013). The number of prospective donors increased yearly over this period, and the prevalence of transfusion-transmissible infections reduced over the 3-year period. Hepatitis B was the most prevalent transfusion-transmissible infections in 2·66% of donors, followed by HIV (1·41%), HCV (0·45%) and VDRL (0·21%). The total prevalence of TTI in this study was 4·73%.

Conclusion Our result shows a steady decrease in the prevalence of TTIs amongst prospective blood donors. This maybe as a result of increase in the number of voluntary unremunerated donors.

Key words: donors, HBV, HCV, HIV, seroprevalence, transfusion-transmissible Infections, VDRL

Introduction

Blood transfusion is a life-saving procedure, but it is not without risks. Some of the risks are trivial and passing, but others may be life-threatening and more persistent. Blood safety is still a significant challenge in Africa [1]; with the HIV/AIDS pandemic, it is observed that both the public and medical practitioners are apprehensive about the safety of blood transfusion services. It is therefore necessary to ensure that safe blood is provided at all times for recipients and the process of donation safe for the donor.

To ensure the safety of blood, stringent donor selection techniques are used – proper organization and recruitment of voluntary unremunerated donors, the use of donor questionnaire and testing of donors for transfusion-transmissible infections (TTI).

There are a number of pathogens which are transmitted by transfusion, but some are considered as the conventional infections that must be tested for, while others are newer infections that have been added to the spectrum, depending on the endemicity, technological advancement and the clinical setting. Given the prevalence of the conventional infective agents in our environment, it is likely that some donors may have these infections and if not properly screened, they may be transmitted to recipients through the blood transfusion. This study was aimed to determine the trends in the seroprevalence of HIV, HBV, HCV and VDRL amongst blood donors in a tertiary referral centre in Nigeria.

Materials and methods

The screening records of 5600 consecutive prospective donors between 2011 and 2013 were retrospectively

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reviewed. Majority of them were family replacement or remunerated donors. The age and sex of the donors, their serological screening test results of the donors were noted.

The serological tests for antibodies to the TTIs were done with rapid detection test kits. Five millilitres of venous blood was collected from each of the donors into a sterile plain bottle, and this was centrifuged to get serum which was separated for analysis. The hepatitis B surface antigen (HBSAg), hepatitis C virus (HCV) antibody and venereal disease research laboratory (VDRL) test for antibodies to *Treponema pallidum* were done with specific DiaSpot © Rapid Detection Testing kits, and HIV screening was done with two kits as recommended by the World Health Organization (WHO) [2]. Determine© HIV-1/2 (Abbott Laboratories, Illinois, IL, USA) and Uni-Gold ©HIV-1/2 (Trinity Biotech, Dublin, Ireland) were used, following the manufacturer's instructions.

Results

A total of 5600 consecutive donor records were reviewed for the 3-year period. They comprised of 5548 (90·32%) male donors and 542 (9·68%) females; their age ranged from 18 to 59 years. Majority of the donors, 2102 (37·54%) were between 28 and 37 years of age (Fig. 1). The number of blood donors increased over the study period (1097–1679 and 2824 per year). Two hundred and seven (3·69%) of these donors were voluntary donors.

The number of voluntary donors increased from 28 in 2011 to 83 in 2012 and 98 in 2013. The total number of donors who were seropositive for one TTI or the other and thus deferred from donation was 265 (4·73%). Although the voluntary donors increased each year, it did not correlate with the trends in the total number of donors each year, $r = 0{\cdot}881$; $P = 0{\cdot}314$.

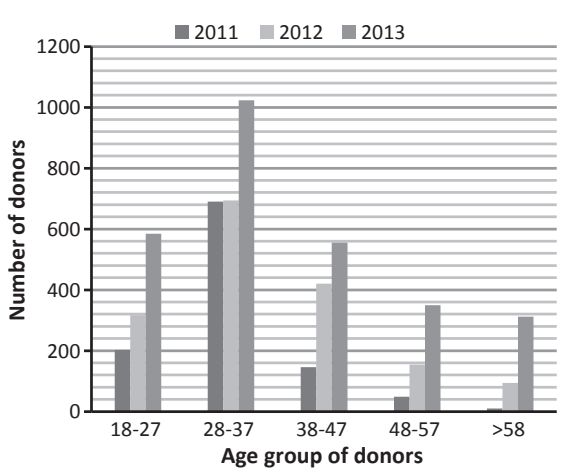


Fig. 1 Age distribution of donors.

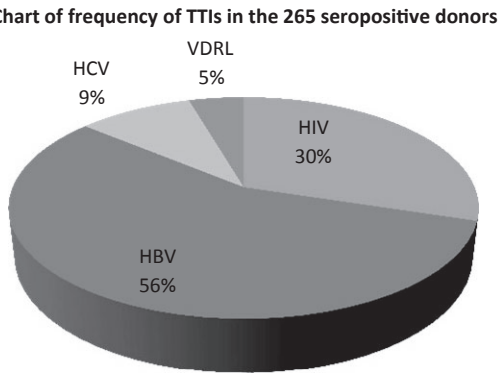


Fig. 2 Pie chart of frequency of TTIs in seropositive donors.

The prevalence of seropositive donors also reduced through the period, from 7·57% in 2011 to 5·36 in 2012 and 3·26% in 2013.

The median age of the donors who were seropositive for TTIs was 32 years. HBSAg positivity was the most prevalent in 149 (56%) of the 265 seropositive donors, followed by HIV in 79 (30%), HCV in 25 (9%) and 12 *T. pallidum* antibody in 5% of the seropositive donors (Fig. 2). The seroprevalence of HBSAg was significantly higher, $\chi^2 = 178{\cdot}027$; $P = <0{\cdot}001$.

The seroprevalence of HIV, HBV, HCV and VDRL was 79 (1·41%), 149 (2·66%), 25 (0·45%) and 12 (0·21%), respectively. Males had more sero-markers of TTIs in this study, 4·38% of male donors (243) and 4·06% of the female donors (22); this was however not statistically significant, $\chi^2 = 0{\cdot}603$; $P = 0{\cdot}437$. Donors within the age group of 28–37 years were more likely to test positive for sero-markers of TTIs (4·23% of the donors in this age group), followed by 2·61% of the donors that were 38–47 years old. Donors that were 18–27 years were less likely to be seropositive as only 1·93% of the donors in this age group were seropositive for TTIs; this was however not statistically significant, $\chi^2 = 2{\cdot}290$; $P = 0{\cdot}130$.

Six donors had coinfections of TTIs (four males and two females between 23–40 years of age), the commonest being the co-infection of HIV and HBV in four of the donors, one donor had HBV–HCV coinfection, and another had HIV and HCV co-infection (Table 1).

Table 1 Table of number of donors and those with TTIs

	Voluntary donors	Total number of donors	HIV + VE	HBV + VE	HCV + VE	VDRL + VE
2011	28	1097	19	48	8	8
2012	83	1679	41	35	12	2
2013	98	2824	19	66	5	2
Total	209	5600	79	149	25	12

Discussion

Blood transfusion is a life-saving procedure; it has been an integral part of medical practice. However, with the HIV/AIDS pandemic, it is observed that both the public and medical practitioners are apprehensive about the safety of blood transfusion service. Thus, transfusion should be made safe for both the recipients and the donors. Prospective blood donors are usually administered the donor questionnaire and screened for TTIs so as to determine who is deferred from donation. However, transfusion is still a source of transmission of some infections.

In many developed countries, HIV, HBV and HCV transmissions through transfusion of blood and blood products have been controlled to a great extent; however, other infections are of concern in transfusion practice such as Chagas, hepatitis E, West Nile Virus, HTLV, CMV, vCJD. Some of these are screened for routinely especially when recipients have transplantable diseases. Apart from the aforementioned techniques to reduce TTIs, nucleic acid testing (NAT) and pathogen inactivation procedures are also considered as important techniques for reducing the risk of TTIs [3].

This study has shown that a total of 265 (4·73%) of the donors in the study period were positive for sero-markers of TTIs; they were deferred from donating blood. Donors between ages 18 and 27 years were less likely to have sero-markers of TTIs; thus, it may be useful to encourage young people in this age group in the community to voluntarily donate blood and be retained as donors if they are serologically negative for TTIs.

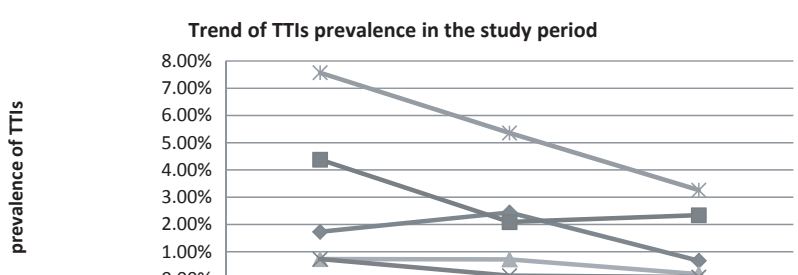
The trend of seropositivity in the study period showed that the percentage of seropositive donors reduced from 7·57% in 2011 to 5·36% in 2012 and 3·26% in 2013 (Fig. 3). This may be associated with increasing number of voluntary donors in the period, although it was not statistically significant. Reports from Eritrea reported an increasing trend despite efforts to prevent TTIs [4]. The study by Salawu *et al.* from Nigeria showed fluctuating trends of individual TTIs [5].

In this study, majority of the donors were positive for hepatitis B surface antigen – 149 (2·66%); the prevalence of HBV fluctuated in this period of study. This was the most prevalent TTI in this study. This is similar to findings from other studies in Nigeria [5–8] and other developing countries like India and Pakistan [9, 10]. It is however different from a study from Bida in west-central Nigeria which had HIV being the most prevalent [11] and Mozambique [12] where syphilis was most prevalent [5].

The prevalence of HBV amongst donors in this study was however lower than reports from other centres in Africa earlier-quoted, but higher than the reports from the Asian countries earlier-quoted [6, 7]. The seroprevalence of hepatitis B in the general population in Port Harcourt (southern Nigeria) is said to be 4·3% [13]; however, a review by Apurva and Jordan [14] put the prevalence in Africa to be 11%. This may contribute to the prevalence of HBV in the donation pool. There have always been more emphases on transmission of HIV through transfusion, but this study shows that there is need for increased awareness on hepatitis B, its modes of transmission and complications in the general population.

The next most prevalent TTI in this study was HIV – 79 (1·41%). This is similar to reports from northern and southern parts of Nigeria [15]; however, it is lower than reports from other parts of Nigeria [2, 4, 8] and Africa [5], but higher than reported for Asia [6, 7]. The trend of seroprevalence of HIV amongst donors fluctuated in this study, similar to findings by Salawu *et al.* [5].

The prevalence of HCV in this study was 25 (0·45%); it is less than the seroprevalence in all the articles quoted.



	2011	2012	2013
HIV	1.73%	2.44%	0.67%
HBV	4.38%	2.09%	2.34%
HCV	0.73%	0.72%	0.18%
VDRL	0.73%	0.12%	0.07%
TOTAL PREVALENCE	7.57%	5.36%	3.26%

Fig. 3 Trends of TTI prevalence in the 3 year period.

However, an earlier report from another health facility in Port Harcourt showed a prevalence of HCV in donors at 5% [16]. HCV seroprevalence showed a consistent downward trend in this study (Fig. 3).

The *Treponema pallidum* antibody was the least prevalent in the study population – 12 (0·21%). This TTI experienced a consistent reduction over the study period. This was similar to the report of Salawu *et al.* Although it is an infrequent cause of TTI, it is a WHO requirement that all donors be screened for *T. pallidum* [17]. Also, the risk of transfusing *T. Pallidum*-infected blood and blood products to pregnant female recipients or their offspring and the attendant complications of congenital syphilis necessitates that blood banks continue to screen for this infection in donors.

TTIs are a major cause of blood transfusion complications, and thus, potential donors that are at risk of transmitting these infections should be excluded by every means possible. Voluntary unremunerated donation from a low-risk population significantly reduces the risk of these infections [18]. It is necessary also to improve the testing methods for these infective agents, from just testing for individuals' immune response to these agents by antibody production using rapid kits, to ELISA-based tests and nucleic acid testing methods so as to shorten the window period and improve the safety of the blood. However, none of these testing methods are 100% sensitive, and they may not be cost-effective for daily use in some of the blood banks in developing countries, considering the workload.

Some limitations of this study are acknowledged; principally that it is a retrospective study of records of blood donors and the seropositive donors were not subjected to further testing to exclude false-positive cases, the seroprevalence of TTIs may have also been overestimated as coinfections were considered separately. In the future, a study on the trends of TTIs in this centre would include ELISA-based testing for TTIs and consecutive donor screening questionnaire records will be properly analysed and reported.

Conclusion

There was a reduced prevalence of sero-markers of TTIs and concomitant increase in the number of donors in this study period, although not statistically significant (P = 0·141). However, the total seroprevalence of TTIs at 4·73% in the study period is not satisfactory and points to the need for increased awareness and improved attempts at reducing these TTIs in the general population, increased requirement of voluntary unremunerated donors, stringent donor selection using standard donor questionnaires and improved testing methods to ensure the safety of transfused blood and blood products.

Disclosure

No conflicts declared by all the authors.

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